Lesson 8: SVD, Norms, and Matrix Calculus

AI/ML Mathematics

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1 Motivation and Overview

How does a neural network learn? At each training step, the algorithm computes the gradient of a scalar loss function with respect to a matrix of weights, updates the weights by a small step in the negative gradient direction, and repeats. Unpacking this single sentence reveals three distinct mathematical needs. First, the weight matrix $W \in \mathbb{R}^{m\times n}$ transforms input features, and understanding its action – which directions it amplifies, which it suppresses, and what its effective rank is – requires the *singular value decomposition* (SVD). Second, bounding the sensitivity of the network output to perturbations in W requires *matrix norms*, and the relationship between the operator norm and the Frobenius norm governs both the stability of the forward pass and the explosion or vanishing of gradients during backpropagation. Third, computing the gradient itself – a derivative of a scalar with respect to a matrix – belongs to *matrix calculus*, the systematic extension of multivariable calculus to matrix-valued arguments.

This capstone lesson of the Linear Algebra subdomain assembles these three pillars: SVD, matrix norms, and matrix calculus. Along the way, we develop quadratic forms (the local shape of any smooth loss near a critical point), the matrix exponential (the bridge to continuous-time dynamics), and the derivative formulas that underpin every gradient computation in deep learning and reinforcement learning.

Where SVD, Norms, and Matrix Calculus Appear in RL/ML

- **PCA and dimensionality reduction.** Principal component analysis computes the top singular vectors of the centered data matrix; the Eckart–Young theorem guarantees this is the best low-rank approximation in both operator and Frobenius norm.
- **Backpropagation and gradient computation.** Every step of backpropagation applies the chain rule for matrix-valued functions; the formulas $\nabla_x(x^\top Ax) = (A + A^\top)x$ and $\partial \operatorname{tr}(AX)/\partial X = A^\top$ appear in nearly every derivation.
- **Condition number and optimization convergence.** The condition number $\kappa(A) = \sigma_1/\sigma_n$ controls the convergence rate of gradient descent; ill-conditioned Hessians cause slow training, motivating preconditioning and adaptive methods like Adam.
- **Policy gradient and natural gradient.** The natural policy gradient requires differentiating the KL-divergence with respect to parameters, which involves quadratic forms in the Fisher information matrix and the matrix calculus chain rule.

Prerequisites. Vector spaces, bases, dimension, linear maps, rank–nullity theorem (Lesson 5). Determinants, eigenvalues, characteristic polynomial, diagonalizability criterion, Cayley–Hamilton theorem (Lesson 6). Inner products, orthonormal bases, Gram–Schmidt, orthogonal projection, spectral theorem for real symmetric matrices, positive definiteness, Rayleigh quotient (Lesson 7).

Learning objectives. After completing this lesson, the student will be able to:

- **Classify** quadratic forms by the eigenvalues of the associated symmetric matrix and connect them to second-order optimality conditions.
- **Prove** the SVD theorem from the spectral decomposition of $A^\top A$ and derive the Eckart–Young low-rank approximation theorem.
- **Compute** the operator norm and Frobenius norm of a matrix in terms of singular values and prove their key relationships.
- **Derive** the matrix exponential and its fundamental properties, including the connection to linear ODEs.

- **Apply** the rules of matrix calculus – gradient, Jacobian, Hessian, trace derivatives, and the chain rule – to compute derivatives arising in optimization and machine learning.
- **Construct** complete gradient derivations for quadratic objectives and trace-based loss functions using the tools of this lesson.

2 References

- Axler [2015, Chapter 7] – Singular value decomposition from the operator-theoretic perspective, polar decomposition.
- Horn and Johnson [2013, Sections 3.1–3.3, 7.4] – Comprehensive treatment of SVD, matrix norms, condition numbers, and the Eckart–Young theorem.
- Strang [2016, Chapters 6–7] – Computational perspective on SVD, least squares, pseudoinverse, and matrix norms.
- Magnus and Neudecker [2019, Chapters 1–6] – Rigorous development of matrix differential calculus with applications.
- Petersen and Pedersen [2012] – Extensive catalog of matrix derivative identities organized by matrix structure.
- Lax [2007, Chapter 6] – Matrix exponential, linear ODEs, and the connection to semigroup theory.

3 Quadratic Forms

3.1 Definition and Canonical Representation

Definition 3.1 (Quadratic Form)**.** Let $A \in \mathbb{R}^{n\times n}$ be a symmetric matrix. The *quadratic form* associated with A is the function $Q_A \colon \mathbb{R}^n \to \mathbb{R}$ defined by

$$Q_A(x) = x^\top A x = \sum_{i=1}^{n}\sum_{j=1}^{n} a_{ij}x_i x_j.$$

Remark 3.2. Any matrix $B \in \mathbb{R}^{n\times n}$ (not necessarily symmetric) gives a quadratic form $x^\top Bx = x^\top\big(\frac{B+B^\top}{2}\big)x$, since $x^\top Bx$ is a scalar and therefore equals its own transpose $x^\top B^\top x$. Hence every quadratic form can be represented by a unique symmetric matrix $A = (B + B^\top)/2$.

Definition 3.3 (Classification of Quadratic Forms)**.** A quadratic form Q_A (equivalently, its symmetric matrix A) is:

- *positive definite* if $Q_A(x) > 0$ for all $x \neq 0$,
- *positive semidefinite* if $Q_A(x) \geq 0$ for all x,
- *negative definite* if $Q_A(x) < 0$ for all $x \neq 0$,
- *negative semidefinite* if $Q_A(x) \leq 0$ for all x,
- *indefinite* if Q_A takes both positive and negative values.

Proposition 3.4 (Eigenvalue Classification)**.** *Let $A \in \mathbb{R}^{n\times n}$ be symmetric with eigenvalues $\lambda_1, \ldots, \lambda_n$. Then:*

1. *$A \succ 0$ if and only if $\lambda_i > 0$ for all i.*
2. *$A \succeq 0$ if and only if $\lambda_i \geq 0$ for all i.*
3. *A is indefinite if and only if A has both positive and negative eigenvalues.*

Proof. By the spectral theorem (Lesson 7, Theorem 5.3), $A = Q\Lambda Q^\top$ where Q is orthogonal and $\Lambda = \mathrm{diag}(\lambda_1, \ldots, \lambda_n)$. Setting $y = Q^\top x$ (a bijection since Q is invertible):

$$Q_A(x) = x^\top Q\Lambda Q^\top x = y^\top \Lambda y = \sum_{i=1}^{n} \lambda_i y_i^2.$$

If all $\lambda_i > 0$, then $\sum_i \lambda_i y_i^2 > 0$ for $y \neq 0$. Conversely, if $\lambda_k \leq 0$ for some k, then $x = q_k$ gives $Q_A(q_k) = \lambda_k \leq 0$. This proves (1). Parts (2) and (3) follow by the same argument. □

Example 3.5 (Classifying a Quadratic Form)**.** Let $A = \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}$. The eigenvalues are $\lambda = 2 \pm 1$, i.e., $\lambda_1 = 1$ and $\lambda_2 = 3$. Since both are positive, $A \succ 0$ and the quadratic form $Q_A(x) = 2x_1^2 - 2x_1x_2 + 2x_2^2$ is positive definite.

3.2 Quadratic Approximation and Second-Order Conditions

Theorem 3.6 (Second-Order Taylor Expansion)**.** *Let $f\colon \mathbb{R}^n \to \mathbb{R}$ be twice continuously differentiable. At a point $x_0 \in \mathbb{R}^n$, for any $h \in \mathbb{R}^n$:*

$$f(x_0 + h) = f(x_0) + \nabla f(x_0)^\top h + \frac{1}{2} h^\top H_f(x_0)\, h + o(\|h\|^2),$$

where $H_f(x_0) \in \mathbb{R}^{n\times n}$ is the Hessian matrix with entries $[H_f(x_0)]_{ij} = \frac{\partial^2 f}{\partial x_i \partial x_j}(x_0)$.

Proof. Apply the one-dimensional Taylor expansion to $g(t) = f(x_0 + th)$ at $t = 0$: $g(1) = g(0) + g'(0) + \frac{1}{2}g''(0) + o(1)$ as $\|h\| \to 0$. By the chain rule, $g'(0) = \nabla f(x_0)^\top h$ and $g''(0) = h^\top H_f(x_0)\, h$. □

Corollary 3.7 (Second-Order Optimality Conditions)**.** *Let $f\colon \mathbb{R}^n \to \mathbb{R}$ be twice continuously differentiable and let x_0 be a critical point ($\nabla f(x_0) = 0$). Then:*

1. *If $H_f(x_0) \succ 0$, then x_0 is a strict local minimum.*
2. *If $H_f(x_0) \prec 0$, then x_0 is a strict local maximum.*
3. *If $H_f(x_0)$ is indefinite, then x_0 is a saddle point.*

Proof. At a critical point, $f(x_0 + h) - f(x_0) = \frac{1}{2}h^\top H_f(x_0)\, h + o(\|h\|^2)$. If $H_f(x_0) \succ 0$, the quadratic term dominates for small $\|h\|$ and is positive for $h \neq 0$, giving a strict local minimum. Similarly for the other cases. □

Example 3.8 (Quadratic Approximation Near a Minimum)**.** Let $f(x_1, x_2) = x_1^4 + x_1^2 + x_2^2$. At $x_0 = (0, 0)$: $\nabla f(0) = (0, 0)^\top$ and $H_f(0) = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} = 2I$. Since $H_f(0) \succ 0$, the origin is a strict local minimum. The quadratic approximation is $f(h) \approx h_1^2 + h_2^2 = \|h\|^2$.

Newton's Method and the Natural Policy Gradient

At a critical point, Newton's method for minimization updates $x \leftarrow x - H_f(x)^{-1}\nabla f(x)$, effectively using the quadratic form $h^\top H_f(x)\, h$ to determine both the direction and step size. In policy optimization, the natural policy gradient replaces the Euclidean Hessian with the Fisher information matrix $F(\theta) \succeq 0$, yielding the update $\theta \leftarrow \theta + \alpha F(\theta)^{-1}\nabla J(\theta)$. The Fisher matrix defines a quadratic form $h^\top F(\theta)\, h$ that measures the local curvature of the KL-divergence between nearby policies. The classification of $F(\theta)$ as positive semidefinite guarantees that this curvature measure is well-defined.

With quadratic forms providing the local geometry of smooth functions, we now turn to a factorization that reveals the global geometry of any linear map: the singular value decomposition.

4 Singular Value Decomposition

4.1 Statement and Proof

Definition 4.1 (Singular Values)**.** Let $A \in \mathbb{R}^{m\times n}$. The *singular values* of A are $\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_{\min(m,n)} \geq 0$, defined as the square roots of the eigenvalues of $A^\top A \in \mathbb{R}^{n\times n}$ (or equivalently of the nonzero eigenvalues of $AA^\top \in \mathbb{R}^{m\times m}$).

Remark 4.2. The matrix $A^\top A$ is symmetric and positive semidefinite, since $x^\top(A^\top A)x = \|Ax\|^2 \geq 0$ for all x. Therefore all eigenvalues of $A^\top A$ are nonneg, and the singular values are well-defined real numbers.

The Singular Value Decomposition

Theorem 4.3 (SVD Theorem)**.** *Let $A \in \mathbb{R}^{m\times n}$ with $\operatorname{rank}(A) = r$. There exist an orthogonal matrix $U \in \mathbb{R}^{m\times m}$, an orthogonal matrix $V \in \mathbb{R}^{n\times n}$, and a matrix $\Sigma \in \mathbb{R}^{m\times n}$ with $\Sigma_{ii} = \sigma_i$ for $i = 1, \ldots, \min(m, n)$ and all other entries zero, such that*

$$A = U\Sigma V^\top.$$

The singular values satisfy $\sigma_1 \geq \cdots \geq \sigma_r > 0 = \sigma_{r+1} = \cdots = \sigma_{\min(m,n)}$. The columns $u_1, \ldots, u_m$ of U are called the left singular vectors*, and the columns $v_1, \ldots, v_n$ of V are called the* right singular vectors*.*

Proof. Since $A^\top A \in \mathbb{R}^{n\times n}$ is symmetric positive semidefinite, the spectral theorem (Lesson 7, Theorem 5.3) provides an orthogonal matrix $V = (v_1 \mid \cdots \mid v_n)$ and eigenvalues $\mu_1 \geq \cdots \geq \mu_n \geq 0$ such that

$$A^\top A = V \operatorname{diag}(\mu_1, \ldots, \mu_n)V^\top.$$

Set $\sigma_i = \sqrt{\mu_i}$ for each i. Since $\operatorname{rank}(A^\top A) = \operatorname{rank}(A) = r$ (the rank of $A^\top A$ equals the rank of A, as $\operatorname{null}(A^\top A) = \operatorname{null}(A)$), we have $\sigma_1 \geq \cdots \geq \sigma_r > 0$ and $\sigma_{r+1} = \cdots = \sigma_n = 0$.

Step 1: Construct the left singular vectors for nonzero singular values. For $i = 1, \ldots, r$, define

$$u_i = \frac{1}{\sigma_i} Av_i.$$

We verify orthonormality: for $1 \leq i, j \leq r$,

$$u_i^\top u_j = \frac{1}{\sigma_i\sigma_j} v_i^\top A^\top Av_j = \frac{1}{\sigma_i\sigma_j} v_i^\top(\sigma_j^2 v_j) = \frac{\sigma_j}{\sigma_i} v_i^\top v_j = \frac{\sigma_j}{\sigma_i}\delta_{ij} = \delta_{ij},$$

where we used $A^\top Av_j = \sigma_j^2 v_j$ and the orthonormality of the v_j.

Step 2: Extend to an orthonormal basis of $\mathbb{R}^m$. The vectors $u_1, \ldots, u_r$ are orthonormal in $\mathbb{R}^m$. If $r < m$, extend them to an orthonormal basis $u_1, \ldots, u_m$ of $\mathbb{R}^m$ (possible by the Gram–Schmidt process applied to any completion). Set $U = (u_1 \mid \cdots \mid u_m)$.

Step 3: Verify $A = U\Sigma V^\top$. It suffices to check $AV = U\Sigma$. For $i = 1, \ldots, r$: $Av_i = \sigma_i u_i = (U\Sigma)_{\cdot,i}$. For $i = r+1, \ldots, n$: $\|Av_i\|^2 = v_i^\top A^\top Av_i = \sigma_i^2\|v_i\|^2 = 0$, so $Av_i = 0 = (U\Sigma)_{\cdot,i}$ (since $\sigma_i = 0$). Thus $AV = U\Sigma$, giving $A = U\Sigma V^\top$. □

4.2 Geometric Interpretation

The SVD reveals that every linear map $A\colon \mathbb{R}^n \to \mathbb{R}^m$ acts by three steps: (1) rotate/reflect by $V^\top$ in the domain, (2) scale by σ_i along each coordinate axis (and embed or project if $m \neq n$), (3) rotate/reflect by U in the codomain. In particular, the image of the unit sphere $\{x : \|x\| = 1\}$ under A is an ellipsoid in $\mathbb{R}^m$ with semi-axes $\sigma_1 u_1, \ldots, \sigma_r u_r$.

Example 4.4 (SVD of a 2×2 Matrix)**.** Let $A = \begin{pmatrix} 3 & 0 \\ 0 & 2 \end{pmatrix}$. Then $A^\top A = \begin{pmatrix} 9 & 0 \\ 0 & 4 \end{pmatrix}$, with eigenvalues $\mu_1 = 9$, $\mu_2 = 4$ and eigenvectors $v_1 = e_1$, $v_2 = e_2$. So $\sigma_1 = 3$, $\sigma_2 = 2$, $u_1 = Av_1/3 = e_1$, $u_2 = Av_2/2 = e_2$. The SVD is $A = I \cdot \mathrm{diag}(3, 2) \cdot I^\top$, which is the matrix itself – diagonal matrices are already in SVD form.

Example 4.5 (SVD of a Non-Square Matrix)**.** Let $A = \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}$. Then $A^\top A = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$, with eigenvalues $\mu_1 = 2$, $\mu_2 = 0$ and eigenvectors $v_1 = \frac{1}{\sqrt{2}}(1, 1)^\top$, $v_2 = \frac{1}{\sqrt{2}}(1, -1)^\top$. So $\sigma_1 = \sqrt{2}$, $\sigma_2 = 0$, and $u_1 = Av_1/\sqrt{2} = (1, 0)^\top = e_1$. Extending: $u_2 = e_2$. The SVD is $A = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} \sqrt{2} & 0 \\ 0 & 0 \end{pmatrix} \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}^\top$. The rank is $r = 1$, confirmed by $\sigma_1 > 0 = \sigma_2$.

Proposition 4.6 (Rank from SVD)**.** $\mathrm{rank}(A)$ *equals the number of nonzero singular values of* A.

Proof. From the SVD, $\mathrm{rank}(A) = \mathrm{rank}(U\Sigma V^\top) = \mathrm{rank}(\Sigma)$ (since U and V are invertible), and $\mathrm{rank}(\Sigma)$ is the number of nonzero diagonal entries. □

4.3 The Eckart–Young Theorem

Best Low-Rank Approximation

Theorem 4.7 (Eckart–Young)**.** *Let $A \in \mathbb{R}^{m\times n}$ have SVD $A = \sum_{i=1}^r \sigma_i u_i v_i^\top$. For any $k \le r$, define $A_k = \sum_{i=1}^k \sigma_i u_i v_i^\top$. Then A_k is the best rank-k approximation to A in both the operator norm and the Frobenius norm:*

$$\min_{\mathrm{rank}(B)\le k} \|A - B\|_{\mathrm{op}} = \|A - A_k\|_{\mathrm{op}} = \sigma_{k+1}, \tag{1}$$

$$\min_{\mathrm{rank}(B)\le k} \|A - B\|_F = \|A - A_k\|_F = \Big(\sum_{i=k+1}^r \sigma_i^2\Big)^{1/2}. \tag{2}$$

(The norms $\|\cdot\|_{\mathrm{op}}$ and $\|\cdot\|_F$ are defined formally in Section 5.)

Proof. We prove the operator norm case; the Frobenius norm case follows by a similar argument.

Achievability: $A - A_k = \sum_{i=k+1}^r \sigma_i u_i v_i^\top$, which has singular values $\sigma_{k+1} \ge \cdots \ge \sigma_r$. Hence $\|A - A_k\|_{\mathrm{op}} = \sigma_{k+1}$ (by Proposition 5.5 below).

Lower bound: Let $B \in \mathbb{R}^{m\times n}$ with $\mathrm{rank}(B) \le k$. Then $\mathrm{null}(B)$ has dimension $\ge n - k$. Let $S = \mathrm{span}(v_1, \ldots, v_{k+1})$, which has dimension $k + 1$. Since $\dim(\mathrm{null}(B)) + \dim(S) \ge (n - k) + (k + 1) = n + 1 > n$, there exists a unit vector $z \in \mathrm{null}(B) \cap S$.

Write $z = \sum_{i=1}^{k+1} c_i v_i$ with $\sum_{i=1}^{k+1} c_i^2 = 1$. Then $Bz = 0$ and

$$\|A - B\|_{\mathrm{op}} \ge \|(A - B)z\| = \|Az\| = \Big\|\sum_{i=1}^{k+1} c_i \sigma_i u_i\Big\| = \Big(\sum_{i=1}^{k+1} c_i^2 \sigma_i^2\Big)^{1/2} \ge \sigma_{k+1},$$

where the last inequality uses $\sigma_i \ge \sigma_{k+1}$ for $i \le k + 1$ and $\sum c_i^2 = 1$.

For the Frobenius norm, a similar dimension argument applied to each singular direction establishes $\|A - B\|_F^2 \ge \sum_{i=k+1}^r \sigma_i^2$ for any rank-k matrix B, with equality at $B = A_k$. □

4.4 The Moore–Penrose Pseudoinverse

Definition 4.8 (Moore–Penrose Pseudoinverse)**.** Let $A \in \mathbb{R}^{m\times n}$ have SVD $A = \sum_{i=1}^{r} \sigma_i u_i v_i^\top$. The *Moore–Penrose pseudoinverse* of A is

$$A^+ = \sum_{i=1}^{r} \frac{1}{\sigma_i} v_i u_i^\top \in \mathbb{R}^{n\times m}.$$

Equivalently, if $A = U\Sigma V^\top$, then $A^+ = V\Sigma^+ U^\top$, where Σ^+ is obtained from Σ by transposing and inverting all nonzero entries.

Theorem 4.9 (Penrose Conditions)**.** *The pseudoinverse A^+ is the unique matrix $X \in \mathbb{R}^{n\times m}$ satisfying:*

1. $AXA = A$,
2. $XAX = X$,
3. $(AX)^\top = AX$,
4. $(XA)^\top = XA$.

Proof. A^+ **satisfies the four conditions.** Using $A = U\Sigma V^\top$ and $X = V\Sigma^+ U^\top$: $AX = U\Sigma\Sigma^+ U^\top$ and $XA = V\Sigma^+\Sigma V^\top$. Both $\Sigma\Sigma^+$ and $\Sigma^+\Sigma$ are diagonal with 1's in the first r positions and 0's elsewhere. Hence AX and XA are symmetric (conditions 3–4), and $AXA = U(\Sigma\Sigma^+\Sigma)V^\top = U\Sigma V^\top = A$ (condition 1), $XAX = V(\Sigma^+\Sigma\Sigma^+)U^\top = V\Sigma^+ U^\top = X$ (condition 2).

Uniqueness. Let X and Y both satisfy conditions 1–4. Then

$$\begin{aligned} X &= XAX = X(AX)^\top = XX^\top A^\top = XX^\top (AYA)^\top = XX^\top A^\top Y^\top A^\top \\ &= (AX)^\top (AY)^\top A^\top = (AX)^\top [(AY)A]^\top = (AX)^\top A^\top Y^\top \\ &= (AAX)^\top Y^\top = (A)^\top Y^\top = (YA)^\top Y^\top \cdot \ ? \end{aligned}$$

A cleaner route: from condition 1, $AXA = A$ and $AYA = A$. Then AX and AY are both orthogonal projections onto range(A) (symmetric and idempotent by conditions 1, 3), so $AX = AY$. Similarly, $XA = YA$ (projections onto range$(A^\top)$). Then $X = XAX = XAY = (XA)Y = (YA)Y = YAY = Y$. □

Theorem 4.10 (Pseudoinverse and Least Squares)**.** *For any $A \in \mathbb{R}^{m\times n}$ and $b \in \mathbb{R}^m$, $x^* = A^+ b$ is the minimum-norm least-squares solution: among all vectors minimizing $\|Ax - b\|$, the vector x^* has the smallest $\|x\|$.*

Proof. The least-squares problem minimizes $\|Ax-b\|^2$, which is equivalent to solving the normal equations $A^\top Ax = A^\top b$. The solution set is $\{x : Ax = \mathrm{proj}_{\mathrm{range}(A)}\, b\} = \{A^+ b + z : z \in \mathrm{null}(A)\}$.

To see that $x^* = A^+ b$ has minimum norm among all solutions, note that $A^+ b \in \mathrm{range}(A^\top) = \mathrm{range}(V_r)$ where $V_r = (v_1 \mid \cdots \mid v_r)$, since $A^+ b = \sum_{i=1}^{r} \frac{u_i^\top b}{\sigma_i} v_i$. Any other solution is $x = x^* + z$ with $z \in \mathrm{null}(A)$. Since $\mathrm{range}(A^\top) \perp \mathrm{null}(A)$ (the fundamental theorem of linear algebra), the Pythagorean theorem gives $\|x\|^2 = \|x^*\|^2 + \|z\|^2 \geq \|x^*\|^2$. □

Example 4.11 (Pseudoinverse of a Rank-1 Matrix)**.** Let $A = \begin{pmatrix} 1 \\ 2 \end{pmatrix} \in \mathbb{R}^{2\times 1}$. The SVD is $A = u_1 \sigma_1 v_1^\top$ with $\sigma_1 = \sqrt{5}$, $u_1 = \frac{1}{\sqrt{5}}(1,2)^\top$, $v_1 = 1$. So $A^+ = v_1 \frac{1}{\sigma_1} u_1^\top = \frac{1}{5}(1,2) = \frac{A^\top}{A^\top A}$. For $b = (3,1)^\top$, the least-squares solution is $x^* = A^+ b = (3+2)/5 = 1$, and $Ax^* = (1,2)^\top$ is the projection of b onto range(A).

SVD in Dimensionality Reduction and Linear Function Approximation

In reinforcement learning with linear function approximation, the value function is approximated as $\hat{V} = \Phi w$ where $\Phi \in \mathbb{R}^{|\mathcal{S}|\times d}$ is a feature matrix. The least-squares solution $w^* = \Phi^+ V^\pi$ uses the pseudoinverse. When d is large, the SVD of Φ reveals which feature directions carry the most information (large singular values) and which are redundant (small singular values). Truncating to the top k singular values gives the best rank-k feature approximation by the Eckart–Young theorem, which is precisely what PCA computes when Φ contains centered data.

With the SVD established, we can now define and analyze the matrix norms that measure the "size" of a linear map.

5 Trace and Matrix Norms

5.1 Trace: Definition and Properties

Definition 5.1 (Trace)**.** The *trace* of a square matrix $A \in \mathbb{R}^{n\times n}$ is $\operatorname{tr}(A) = \sum_{i=1}^n a_{ii}$.

Proposition 5.2 (Properties of the Trace)**.** *For matrices $A, B \in \mathbb{R}^{n\times n}$, $C \in \mathbb{R}^{m\times n}$, $D \in \mathbb{R}^{n\times m}$, and $\alpha \in \mathbb{R}$:*

1. ***Linearity:*** $\operatorname{tr}(\alpha A + B) = \alpha \operatorname{tr}(A) + \operatorname{tr}(B)$.
2. ***Cyclic property:*** $\operatorname{tr}(CD) = \operatorname{tr}(DC)$.
3. ***Similarity invariance:*** *If P is invertible, then* $\operatorname{tr}(P^{-1}AP) = \operatorname{tr}(A)$.
4. ***Eigenvalue sum:*** $\operatorname{tr}(A) = \sum_{i=1}^n \lambda_i$ *where λ_i are the eigenvalues of A (counted with algebraic multiplicity).*
5. ***Frobenius inner product:*** $\operatorname{tr}(A^\top B) = \sum_{i,j} a_{ij} b_{ij}$.

Proof. (1) Direct from the definition: $\operatorname{tr}(\alpha A + B) = \sum_i (\alpha a_{ii} + b_{ii}) = \alpha \sum_i a_{ii} + \sum_i b_{ii}$.

(2) Let $C = (c_{ik}) \in \mathbb{R}^{m\times n}$, $D = (d_{kj}) \in \mathbb{R}^{n\times m}$. Then $\operatorname{tr}(CD) = \sum_{i=1}^m \sum_{k=1}^n c_{ik} d_{ki} = \sum_{k=1}^n \sum_{i=1}^m d_{ki} c_{ik} = \operatorname{tr}(DC)$.

(3) Follows from (2): $\operatorname{tr}(P^{-1}AP) = \operatorname{tr}(APP^{-1}) = \operatorname{tr}(A)$.

(4) Over $\mathbb{C}$, every matrix is similar to an upper triangular matrix T (Schur decomposition; see Lesson 6). Then $\operatorname{tr}(A) = \operatorname{tr}(T) = \sum_i t_{ii} = \sum_i \lambda_i$.

(5) $\operatorname{tr}(A^\top B) = \sum_i (A^\top B)_{ii} = \sum_i \sum_j a_{ji} b_{ji} = \sum_{i,j} a_{ij} b_{ij}$. □

Example 5.3 (Trace as an Inner Product on Matrices)**.** The *Frobenius inner product* on $\mathbb{R}^{m\times n}$ is $\langle A, B\rangle_F = \operatorname{tr}(A^\top B)$. It satisfies all inner product axioms: symmetry ($\operatorname{tr}(A^\top B) = \operatorname{tr}(B^\top A)$), bilinearity (from linearity of trace), and positive definiteness ($\operatorname{tr}(A^\top A) = \sum_{ij} a_{ij}^2 \geq 0$, with equality iff $A = 0$). This turns $\mathbb{R}^{m\times n}$ into an inner product space of dimension mn.

5.2 Operator Norm

Definition 5.4 (Operator Norm)**.** For $A \in \mathbb{R}^{m\times n}$, the *operator norm* (or *spectral norm*, or *induced 2-norm*) is

$$\|A\|_{\text{op}} = \max_{\|x\|=1} \|Ax\| = \max_{x\neq 0} \frac{\|Ax\|}{\|x\|},$$

where $\|\cdot\|$ denotes the Euclidean norm on $\mathbb{R}^n$ and $\mathbb{R}^m$.

Proposition 5.5 (Operator Norm Equals σ_1)**.** $\|A\|_{\text{op}} = \sigma_1(A)$*, the largest singular value of A.*

Proof. For any unit vector x, write $x = \sum_j c_j v_j$ with $\sum_j c_j^2 = 1$ (in the right singular vector basis). Then $Ax = \sum_{j=1}^r c_j \sigma_j u_j$ and $\|Ax\|^2 = \sum_{j=1}^r c_j^2 \sigma_j^2 \leq \sigma_1^2 \sum_j c_j^2 = \sigma_1^2$. Equality holds at $x = v_1$. □

5.3 Frobenius Norm

Definition 5.6 (Frobenius Norm)**.** The *Frobenius norm* of $A \in \mathbb{R}^{m\times n}$ is

$$\|A\|_F = \sqrt{\sum_{i=1}^{m}\sum_{j=1}^{n} a_{ij}^2} = \sqrt{\operatorname{tr}(A^\top A)}.$$

Proposition 5.7 (Frobenius Norm from Singular Values)**.** $\|A\|_F = \sqrt{\sigma_1^2 + \cdots + \sigma_r^2}$ *where* $\sigma_1, \ldots, \sigma_r$ *are the nonzero singular values of* A*.*

Proof. $\|A\|_F^2 = \operatorname{tr}(A^\top A) = \operatorname{tr}(V\Sigma^\top\Sigma V^\top) = \operatorname{tr}(\Sigma^\top\Sigma) = \sum_i \sigma_i^2$, using the cyclic property of trace and the fact that $\Sigma^\top\Sigma = \operatorname{diag}(\sigma_1^2, \ldots, \sigma_n^2)$. □

Example 5.8 (Computing Both Norms)**.** Let $A = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}$. The Frobenius norm is $\|A\|_F = \sqrt{1+4+0+1} = \sqrt{6}$. For the operator norm, $A^\top A = \begin{pmatrix} 1 & 2 \\ 2 & 5 \end{pmatrix}$ with eigenvalues $\mu = 3 \pm \sqrt{5}$. So $\sigma_1 = \sqrt{3+\sqrt{5}} \approx 2.29$ and $\|A\|_{\text{op}} = \sigma_1$. We verify: $\sigma_1^2 + \sigma_2^2 = (3+\sqrt{5}) + (3-\sqrt{5}) = 6 = \|A\|_F^2$.

5.4 Norm Relationships

Theorem 5.9 (Norm Inequalities)**.** *For* $A \in \mathbb{R}^{m\times n}$ *with* $\operatorname{rank}(A) = r$*:*

$$\|A\|_{\text{op}} \le \|A\|_F \le \sqrt{r}\,\|A\|_{\text{op}}.$$

Proof. For the left inequality: $\|A\|_{\text{op}}^2 = \sigma_1^2 \le \sigma_1^2 + \cdots + \sigma_r^2 = \|A\|_F^2$.

For the right inequality: $\|A\|_F^2 = \sum_{i=1}^r \sigma_i^2 \le r \cdot \sigma_1^2 = r \cdot \|A\|_{\text{op}}^2$.

Both bounds are tight: the left is tight when $\operatorname{rank}(A) = 1$ (all but one singular value is zero), and the right is tight when $\sigma_1 = \cdots = \sigma_r$ (e.g., $A = I_r$ padded with zeros). □

5.5 Submultiplicativity

Theorem 5.10 (Submultiplicativity)**.** *For* $A \in \mathbb{R}^{m\times p}$ *and* $B \in \mathbb{R}^{p\times n}$*:*

1. $\|AB\|_{\text{op}} \le \|A\|_{\text{op}} \cdot \|B\|_{\text{op}}$.
2. $\|AB\|_F \le \|A\|_{\text{op}} \cdot \|B\|_F$.
3. $\|AB\|_F \le \|A\|_F \cdot \|B\|_F$.

Proof. (1) For any unit vector x: $\|ABx\| \le \|A\|_{\text{op}}\|Bx\| \le \|A\|_{\text{op}}\|B\|_{\text{op}}\|x\| = \|A\|_{\text{op}}\|B\|_{\text{op}}$. Taking the maximum over $\|x\| = 1$ gives the result.

(2) Let $b_1, \ldots, b_n$ be the columns of B. Then $\|AB\|_F^2 = \sum_{j=1}^n \|Ab_j\|^2 \le \|A\|_{\text{op}}^2 \sum_{j=1}^n \|b_j\|^2 = \|A\|_{\text{op}}^2\|B\|_F^2$.

(3) By the Cauchy–Schwarz inequality on the Frobenius inner product: $(AB)_{ij} = \sum_k a_{ik}b_{kj}$, so $\|AB\|_F^2 = \sum_{i,j}\left(\sum_k a_{ik}b_{kj}\right)^2$. Alternatively, from (2): $\|AB\|_F \le \|A\|_{\text{op}}\|B\|_F \le \|A\|_F\|B\|_F$ using $\|A\|_{\text{op}} \le \|A\|_F$ (Theorem 5.9). □

5.6 Condition Number

Definition 5.11 (Condition Number)**.** For an invertible matrix $A \in \mathbb{R}^{n\times n}$, the *condition number* (with respect to the operator norm) is

$$\kappa(A) = \|A\|_{\text{op}} \cdot \|A^{-1}\|_{\text{op}} = \frac{\sigma_1(A)}{\sigma_n(A)}.$$

Theorem 5.12 (Sensitivity of Linear Systems). *Let $A \in \mathbb{R}^{n\times n}$ be invertible and let $Ax = b$. If b is perturbed to $b + \delta b$ with solution $x + \delta x$, then*

$$\frac{\|\delta x\|}{\|x\|} \le \kappa(A)\,\frac{\|\delta b\|}{\|b\|}.$$

Proof. From $A(x+\delta x) = b+\delta b$, we get $A\,\delta x = \delta b$, so $\delta x = A^{-1}\delta b$. Hence $\|\delta x\| \le \|A^{-1}\|_{\mathrm{op}}\|\delta b\|$. Also $\|b\| = \|Ax\| \le \|A\|_{\mathrm{op}}\|x\|$, so $1/\|x\| \le \|A\|_{\mathrm{op}}/\|b\|$. Combining:

$$\frac{\|\delta x\|}{\|x\|} \le \|A^{-1}\|_{\mathrm{op}}\|A\|_{\mathrm{op}}\frac{\|\delta b\|}{\|b\|} = \kappa(A)\frac{\|\delta b\|}{\|b\|}.$$ □

Example 5.13 (Condition Number and Sensitivity). The matrix $A = \begin{pmatrix} 1 & 0 \\ 0 & 0.001 \end{pmatrix}$ has $\sigma_1 = 1$, $\sigma_2 = 0.001$, so $\kappa(A) = 1000$. A 0.1% perturbation in b can cause up to a 100% change in x. In contrast, $A = I$ has $\kappa = 1$ (perfectly conditioned).

Condition Number and Numerical Stability

The condition number $\kappa(A)$ is an intrinsic property of the problem $Ax = b$, not of any particular algorithm. A large condition number means the problem is inherently sensitive to perturbations. In machine learning, the condition number of the Hessian $\kappa(H_f)$ determines the convergence rate of gradient descent: the number of iterations scales as $O(\kappa \log(1/\varepsilon))$. This motivates preconditioning techniques and adaptive methods that implicitly rescale the problem to reduce κ.

With norms providing quantitative measures of matrix size and sensitivity, we now develop the matrix exponential, which connects linear algebra to continuous-time dynamics.

6 Matrix Exponential

6.1 Definition and Convergence

Definition 6.1 (Matrix Exponential). For $A \in \mathbb{R}^{n\times n}$, the *matrix exponential* is

$$e^A = \exp(A) = \sum_{k=0}^{\infty} \frac{A^k}{k!} = I + A + \frac{A^2}{2!} + \frac{A^3}{3!} + \cdots.$$

Proposition 6.2 (Convergence of the Matrix Exponential). *The series $\sum_{k=0}^{\infty} A^k/k!$ converges absolutely (in any matrix norm) for every $A \in \mathbb{R}^{n\times n}$.*

Proof. Using submultiplicativity (Theorem 5.10), $\|A^k/k!\|_{\mathrm{op}} \le \|A\|_{\mathrm{op}}^k/k!$. The scalar series $\sum_k \|A\|_{\mathrm{op}}^k/k! = e^{\|A\|_{\mathrm{op}}} < \infty$ converges, so the matrix series converges absolutely. (Absolute convergence in a complete normed space implies convergence.) □

6.2 Properties

Theorem 6.3 (Properties of the Matrix Exponential). *Let $A, B \in \mathbb{R}^{n\times n}$.*

1. $e^0 = I$.
2. *If $AB = BA$, then $e^{A+B} = e^A e^B$.*
3. *e^A is always invertible, with $(e^A)^{-1} = e^{-A}$.*
4. $\frac{d}{dt}e^{tA} = Ae^{tA} = e^{tA}A$.
5. *If $A = PDP^{-1}$ (diagonalizable), then $e^A = P\,\mathrm{diag}(e^{\lambda_1}, \ldots, e^{\lambda_n})\,P^{-1}$.*

Proof. (1) $e^0 = \sum_k 0^k/k! = I$ (where $0^0 = I$ by convention).

(2) When $AB = BA$, the binomial theorem applies: $(A+B)^k = \sum_{j=0}^{k} \binom{k}{j} A^j B^{k-j}$. Then

$$e^{A+B} = \sum_{k=0}^{\infty} \frac{(A+B)^k}{k!} = \sum_{k=0}^{\infty} \sum_{j=0}^{k} \frac{A^j}{j!} \frac{B^{k-j}}{(k-j)!} = \Big(\sum_{j=0}^{\infty} \frac{A^j}{j!}\Big)\Big(\sum_{\ell=0}^{\infty} \frac{B^\ell}{\ell!}\Big) = e^A e^B,$$

where the rearrangement is justified by absolute convergence.

(3) Since A commutes with $-A$, part (2) gives $e^A e^{-A} = e^{A-A} = e^0 = I$.

(4) Differentiating term by term (justified by uniform convergence on compact intervals):

$$\frac{d}{dt} e^{tA} = \frac{d}{dt} \sum_{k=0}^{\infty} \frac{t^k A^k}{k!} = \sum_{k=1}^{\infty} \frac{t^{k-1} A^k}{(k-1)!} = A \sum_{k=1}^{\infty} \frac{t^{k-1} A^{k-1}}{(k-1)!} = A e^{tA}.$$

The identity $e^{tA} A$ follows since A commutes with e^{tA} (each term $t^k A^k/k!$ commutes with A).

(5) $A^k = PD^kP^{-1}$, so $e^A = P\big(\sum_k D^k/k!\big)P^{-1} = P \operatorname{diag}(e^{\lambda_1}, \dots, e^{\lambda_n})\, P^{-1}$. □

Example 6.4 (Matrix Exponential of a Diagonal Matrix)**.** If $A = \operatorname{diag}(-1, 2)$, then $e^{tA} = \operatorname{diag}(e^{-t}, e^{2t})$. As $t \to \infty$, the first component decays to 0 and the second grows without bound. The eigenvalues $\{-1, 2\}$ determine the long-time behavior.

Example 6.5 (Matrix Exponential and Rotation)**.** Let $A = \begin{pmatrix} 0 & -\theta \\ \theta & 0 \end{pmatrix}$ for $\theta \in \mathbb{R}$. Since A is skew-symmetric ($A^\top = -A$), e^A is orthogonal. Using $A^2 = -\theta^2 I$, the series groups as

$$e^A = I \cos\theta + \frac{A}{\theta} \sin\theta = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix},$$

which is a rotation by angle θ.

Example 6.6 (Linear ODE Solution)**.** The system $\dot{x}(t) = Ax(t)$ with initial condition $x(0) = x_0$ has the unique solution $x(t) = e^{tA} x_0$. This follows from property (4): $\frac{d}{dt}(e^{tA} x_0) = A e^{tA} x_0 = Ax(t)$. In continuous-time Markov chains, the transition rate matrix Q (with $q_{ij} \geq 0$ for $i \neq j$ and row sums zero) gives the transition probability matrix $P(t) = e^{tQ}$.

We now develop the final topic: the systematic computation of derivatives of matrix-valued expressions.

7 Matrix Calculus

7.1 Gradient, Jacobian, and Hessian

Definition 7.1 (Gradient)**.** Let $f\colon \mathbb{R}^n \to \mathbb{R}$ be differentiable at x. The *gradient* of f at x is the unique vector $\nabla f(x) \in \mathbb{R}^n$ satisfying

$$f(x+h) = f(x) + \nabla f(x)^\top h + o(\|h\|) \quad \text{as } \|h\| \to 0.$$

In coordinates, $[\nabla f(x)]_i = \partial f / \partial x_i$.

Definition 7.2 (Jacobian)**.** Let $g\colon \mathbb{R}^n \to \mathbb{R}^m$ be differentiable at x. The *Jacobian* of g at x is the matrix $J_g(x) \in \mathbb{R}^{m \times n}$ with $[J_g(x)]_{ij} = \partial g_i / \partial x_j$, satisfying

$$g(x+h) = g(x) + J_g(x)\, h + o(\|h\|).$$

When $m = 1$, $J_g(x) = \nabla g(x)^\top$.

Definition 7.3 (Hessian)**.** Let $f\colon \mathbb{R}^n \to \mathbb{R}$ be twice differentiable at x. The *Hessian* of f at x is $H_f(x) \in \mathbb{R}^{n \times n}$ with $[H_f(x)]_{ij} = \frac{\partial^2 f}{\partial x_i \partial x_j}$. If the second partial derivatives are continuous, $H_f(x)$ is symmetric. The Hessian is the Jacobian of the gradient: $H_f(x) = J_{\nabla f}(x)$.

7.2 Fundamental Vector Derivative Formulas

Theorem 7.4 (Basic Gradient Formulas)**.** *Let $a \in \mathbb{R}^n$, $A \in \mathbb{R}^{n\times n}$, and $x \in \mathbb{R}^n$.*

1. $\nabla_x(a^\top x) = a$.
2. $\nabla_x(x^\top Ax) = (A + A^\top)x$. *If A is symmetric, this simplifies to $2Ax$.*
3. $\nabla_x(\|x\|^2) = 2x$.
4. $\nabla_x(\|Ax - b\|^2) = 2A^\top(Ax - b)$ *for $b \in \mathbb{R}^m$, $A \in \mathbb{R}^{m\times n}$.*

Proof. (1) $f(x+h) = a^\top(x+h) = a^\top x+a^\top h = f(x)+a^\top h$. Comparing with $f(x)+\nabla f(x)^\top h+o(\|h\|)$ gives $\nabla f(x) = a$.

(2) $f(x + h) = (x + h)^\top A(x + h) = x^\top Ax + x^\top Ah + h^\top Ax + h^\top Ah$. The linear term in h is $x^\top Ah + h^\top Ax = (A^\top x)^\top h + (Ax)^\top h = (A^\top x + Ax)^\top h = [(A + A^\top)x]^\top h$. The term $h^\top Ah$ is $o(\|h\|)$. Hence $\nabla f(x) = (A + A^\top)x$.

(3) $\|x\|^2 = x^\top Ix$. Apply (2) with $A = I$: $\nabla_x(x^\top x) = 2Ix = 2x$.

(4) Set $g(x) = \|Ax - b\|^2 = (Ax - b)^\top(Ax - b) = x^\top A^\top Ax - 2b^\top Ax + b^\top b$. By (1) and (2): $\nabla g(x) = (A^\top A + A^\top A)x - 2A^\top b = 2A^\top Ax - 2A^\top b = 2A^\top(Ax - b)$. □

Example 7.5 (Gradient of a Quadratic Loss)**.** The least-squares loss $L(w) = \frac{1}{2}\|Xw - y\|^2$ with $X \in \mathbb{R}^{m\times n}$ and $y \in \mathbb{R}^m$ has gradient $\nabla L(w) = X^\top(Xw - y)$ (by Theorem 7.4(4) with a factor of 1/2). Setting $\nabla L = 0$ gives the normal equations $X^\top Xw = X^\top y$, whose minimum-norm solution is $w^* = X^+y$.

Example 7.6 (Gradient of a Regularized Loss)**.** Ridge regression adds an ℓ_2 penalty: $L(w) = \frac{1}{2}\|Xw - y\|^2 + \frac{\lambda}{2}\|w\|^2$. Using Theorem 7.4: $\nabla L(w) = X^\top(Xw - y) + \lambda w$. Setting to zero: $(X^\top X + \lambda I)w = X^\top y$, so $w^* = (X^\top X + \lambda I)^{-1}X^\top y$. The regularization term λI improves the condition number of $X^\top X$.

7.3 Trace Derivative Formulas

When the variable is a matrix $X \in \mathbb{R}^{m\times n}$, the gradient $\nabla_X f$ is the matrix of partial derivatives $[\nabla_X f]_{ij} = \partial f/\partial x_{ij}$, characterized by

$$f(X + H) = f(X) + \operatorname{tr}(\nabla_X f(X)^\top H) + o(\|H\|_F).$$

Theorem 7.7 (Trace Derivative Formulas)**.** *Let $A \in \mathbb{R}^{n\times m}$, $B \in \mathbb{R}^{m\times n}$, $X \in \mathbb{R}^{m\times n}$.*

1. $\frac{\partial}{\partial X}\operatorname{tr}(AX) = A^\top$.
2. $\frac{\partial}{\partial X}\operatorname{tr}(X^\top A) = A$.
3. $\frac{\partial}{\partial X}\operatorname{tr}(X^\top AX) = (A + A^\top)X$ *(for $X \in \mathbb{R}^{n\times p}$, $A \in \mathbb{R}^{n\times n}$).*
4. $\frac{\partial}{\partial X}\operatorname{tr}(AXB) = A^\top B^\top$ *(for $A \in \mathbb{R}^{p\times m}$, $B \in \mathbb{R}^{n\times p}$).*

Proof. (1) $\operatorname{tr}(A(X + H)) = \operatorname{tr}(AX) + \operatorname{tr}(AH)$. Since $\operatorname{tr}(AH) = \operatorname{tr}(A^{\top\top}H) = \sum_{ij}[A^\top]_{ij}h_{ij} = \operatorname{tr}((A^\top)^\top H)$, the gradient is $A^\top$.

(2) $\operatorname{tr}((X + H)^\top A) = \operatorname{tr}(X^\top A) + \operatorname{tr}(H^\top A)$. Now $\operatorname{tr}(H^\top A) = \sum_{ij} h_{ij}a_{ij} = \operatorname{tr}(A^\top H)$, so the gradient is A.

(3) $\operatorname{tr}((X + H)^\top A(X + H)) = \operatorname{tr}(X^\top AX) + \operatorname{tr}(H^\top AX) + \operatorname{tr}(X^\top AH) + \operatorname{tr}(H^\top AH)$. The linear part is $\operatorname{tr}(H^\top AX)+\operatorname{tr}(X^\top AH) = \operatorname{tr}(H^\top AX)+\operatorname{tr}(H^\top A^\top X) = \operatorname{tr}(H^\top(A+A^\top)X)$, giving gradient $(A + A^\top)X$.

(4) $\operatorname{tr}(A(X + H)B) = \operatorname{tr}(AXB) + \operatorname{tr}(AHB)$. Using the cyclic property: $\operatorname{tr}(AHB) = \operatorname{tr}(BAHB \cdot B^{-1}) = \operatorname{tr}(BAH)$ – more directly, $\operatorname{tr}(AHB) = \operatorname{tr}(BAH) = \operatorname{tr}((A^\top B^\top)^\top H)$, so the gradient is $A^\top B^\top$. □

Example 7.8 (Gradient of $\|AX - B\|_F^2$)**.** Let $f(X) = \|AX - B\|_F^2 = \operatorname{tr}((AX - B)^\top(AX - B))$. Expanding: $f(X) = \operatorname{tr}(X^\top A^\top AX) - 2\operatorname{tr}(B^\top AX) + \operatorname{tr}(B^\top B)$. Using Theorem 7.7: $\nabla_X f = 2A^\top AX - 2A^\top B = 2A^\top(AX - B)$. This extends the vector least-squares gradient to the matrix case.

7.4 The Chain Rule

Theorem 7.9 (Chain Rule for Compositions)**.** *Let $g\colon \mathbb{R}^n \to \mathbb{R}^m$ be differentiable at x and $f\colon \mathbb{R}^m \to \mathbb{R}$ be differentiable at $g(x)$. Then the composition $h = f \circ g\colon \mathbb{R}^n \to \mathbb{R}$ is differentiable at x with*

$$\nabla h(x) = J_g(x)^\top \nabla f(g(x)).$$

More generally, if $f\colon \mathbb{R}^m \to \mathbb{R}^p$, then $J_{f\circ g}(x) = J_f(g(x))\, J_g(x)$.

Proof. By differentiability of g and f:

$$\begin{aligned} h(x+\delta) &= f(g(x+\delta)) = f(g(x) + J_g(x)\delta + o(\|\delta\|)) \\ &= f(g(x)) + \nabla f(g(x))^\top [J_g(x)\delta + o(\|\delta\|)] + o(\|J_g(x)\delta + o(\|\delta\|)\|) \\ &= f(g(x)) + \nabla f(g(x))^\top J_g(x)\delta + o(\|\delta\|) \\ &= h(x) + [J_g(x)^\top \nabla f(g(x))]^\top \delta + o(\|\delta\|). \end{aligned}$$

Hence $\nabla h(x) = J_g(x)^\top \nabla f(g(x))$. The general case follows by applying this to each component of f. □

Matrix Calculus Toolbox for Gradient Computation

The results of this section provide the complete toolbox for computing gradients in machine learning:

1. **Vector derivatives:** $\nabla(a^\top x) = a$, $\nabla(x^\top A x) = (A + A^\top)x$ (Theorem 7.4).
2. **Trace derivatives:** $\partial \operatorname{tr}(AX)/\partial X = A^\top$, $\partial \operatorname{tr}(X^\top A X)/\partial X = (A + A^\top)X$ (Theorem 7.7).
3. **Chain rule:** $\nabla(f \circ g)(x) = J_g(x)^\top \nabla f(g(x))$ (Theorem 7.9).

Every gradient computation in backpropagation is an iterated application of these three rules. The chain rule composes the Jacobians of individual layers, while the vector and trace formulas compute the Jacobian of each layer.

Example 7.10 (Softmax Cross-Entropy Gradient)**.** In a classification network, the final layer computes logits $z = Wx + b \in \mathbb{R}^K$, then $p = \operatorname{softmax}(z)$ with $p_k = e^{z_k}/\sum_j e^{z_j}$, and the loss is $L = -\log p_y$ (cross-entropy for true class y). By the chain rule: $\nabla_z L = p - e_y$ (where e_y is the y-th standard basis vector), and then $\nabla_W L = (p - e_y)x^\top$ (a rank-1 outer product). Each step uses the chain rule (Theorem 7.9) applied to the composition of the linear layer, softmax, and log-loss.

8 Exercises

Exercise 8.1 (Quadratic Form Classification)**.** (∗)
For each matrix, classify the associated quadratic form as positive definite, positive semidefinite, negative definite, negative semidefinite, or indefinite:

(a) $A = \begin{pmatrix} 5 & 2 \\ 2 & 1 \end{pmatrix}$.

(b) $B = \begin{pmatrix} -3 & 0 \\ 0 & -1 \end{pmatrix}$.

(c) $C = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix}$.

Exercise 8.2 (SVD Computation). $(*)$
Compute the full SVD of $A = \begin{pmatrix} 1 & 1 \\ 1 & -1 \\ 0 & 0 \end{pmatrix}$.

(a) Find $A^\top A$ and its eigenvalues and eigenvectors.
(b) Determine the singular values and right singular vectors.
(c) Construct the left singular vectors and verify $A = U\Sigma V^\top$.
(d) Find the pseudoinverse A^+ and verify $AA^+A = A$.

Exercise 8.3 (Low-Rank Approximation in RL). $(**)$
In a reinforcement learning problem with $|\mathcal{S}| = 100$ states and $d = 10$ features, the feature matrix $\Phi \in \mathbb{R}^{100\times 10}$ has singular values $\sigma_1 = 5, \sigma_2 = 4, \sigma_3 = 3, \sigma_4 = 2, \sigma_5 = 1, \sigma_6 = \cdots = \sigma_{10} = 0.01$.

(a) Compute $\kappa(\Phi^\top\Phi)$ and interpret its meaning for the least-squares problem $\Phi w \approx V^\pi$.
(b) If we truncate Φ to rank $k = 5$, what is the Frobenius norm of the approximation error $\|\Phi - \Phi_5\|_F$?
(c) What fraction of $\|\Phi\|_F^2$ is captured by the rank-5 approximation?

Exercise 8.4 (Matrix Exponential Computation). $(**)$
Let $A = \begin{pmatrix} 1 & 1 \\ 0 & 2 \end{pmatrix}$.

(a) Find P and D such that $A = PDP^{-1}$.
(b) Compute e^{tA} for all $t \in \mathbb{R}$.
(c) Verify that $\frac{d}{dt}e^{tA}\big|_{t=0} = A$.
(d) Solve the ODE $\dot{x} = Ax$ with $x(0) = (1, 0)^\top$.

Exercise 8.5 (Trace Derivative Derivation). $(**)$

(a) Prove that $\frac{\partial}{\partial X}\operatorname{tr}(AX^\top BX) = A^\top XB^\top + BXA$ for $X \in \mathbb{R}^{m\times n}$, $A \in \mathbb{R}^{m\times m}$, $B \in \mathbb{R}^{n\times n}$.
(b) Using part (a) and the identity $\|AX-B\|_F^2 = \operatorname{tr}((AX-B)^\top(AX-B))$, re-derive $\nabla_X\|AX-B\|_F^2 = 2A^\top(AX-B)$.

Hint: For (a), expand $\operatorname{tr}(A(X+H)^\top B(X+H))$ and isolate the terms linear in H.

Exercise 8.6 (Gradient of the Log-Determinant). $(***)$
Let $f(X) = \log\det(X)$ for symmetric positive definite $X \in \mathbb{R}^{n\times n}$.

(a) Show that $\det(X + tH) = \det(X)\det(I + tX^{-1}H)$ for small t.
(b) Using $\det(I + tM) = 1 + t\operatorname{tr}(M) + O(t^2)$, prove that $\frac{d}{dt}\big|_{t=0}\log\det(X + tH) = \operatorname{tr}(X^{-1}H)$.
(c) Conclude that $\nabla_X \log\det(X) = X^{-1}$ (in the sense of the Frobenius inner product: $\operatorname{tr}((\nabla f)^\top H) = \operatorname{tr}(X^{-1}H)$).
(d) Discuss the relevance of this result to Gaussian distributions, where the log-likelihood involves $\log\det(\Sigma)$ for the covariance matrix Σ.

Hint: For (b), recall that $\log(1 + \varepsilon) = \varepsilon + O(\varepsilon^2)$.

9 Summary

- A *quadratic form* $Q_A(x) = x^\top Ax$ is classified by the eigenvalues of the symmetric matrix A: positive definite iff all $\lambda_i > 0$, positive semidefinite iff all $\lambda_i \geq 0$, indefinite iff A has eigenvalues of both signs. At a critical point, the Hessian quadratic form determines the local nature of the extremum.

- The *singular value decomposition* $A = U\Sigma V^\top$ factors any $m \times n$ matrix into orthogonal rotations and a diagonal scaling. The singular values are the square roots of the eigenvalues of $A^\top A$, and $\operatorname{rank}(A)$ equals the number of nonzero singular values.

- The *Eckart–Young theorem* guarantees that truncating the SVD at rank k gives the best rank-k approximation in both operator and Frobenius norm. This is the theoretical foundation of PCA.
- The *Moore–Penrose pseudoinverse* A^+ is the unique matrix satisfying the four Penrose conditions. The vector A^+b is the minimum-norm least-squares solution to $Ax = b$.
- The *operator norm* $\|A\|_{\mathrm{op}} = \sigma_1$ and the *Frobenius norm* $\|A\|_F = (\sum \sigma_i^2)^{1/2}$ satisfy $\|A\|_{\mathrm{op}} \leq \|A\|_F \leq \sqrt{r}\,\|A\|_{\mathrm{op}}$. Both are submultiplicative.
- The *condition number* $\kappa(A) = \sigma_1/\sigma_n$ bounds the relative amplification of perturbations in linear systems. Large κ means the system is ill-conditioned.
- The *matrix exponential* $e^A = \sum_{k=0}^{\infty} A^k/k!$ converges for every square matrix. It satisfies $\frac{d}{dt}e^{tA} = Ae^{tA}$, solving the linear ODE $\dot{x} = Ax$ with $x(t) = e^{tA}x_0$.
- *Matrix calculus* provides systematic rules: $\nabla(a^\top x) = a$, $\nabla(x^\top Ax) = (A+A^\top)x$, $\partial\,\mathrm{tr}(AX)/\partial X = A^\top$, and the chain rule $\nabla(f \circ g) = J_g^\top \nabla f$.
- The trace function defines an inner product on matrices ($\langle A, B\rangle_F = \mathrm{tr}(A^\top B)$) and provides a unified framework for expressing matrix derivatives.

Lesson	Title	Key Results
5	Vector Spaces and Linear Maps	Vector space axioms, subspaces, bases and dimension, linear maps, rank–nullity theorem, matrix representation, change of basis
6	Determinants and Eigenvalues	Determinant (cofactor, Leibniz), multiplicativity, eigenvalues, characteristic polynomial, algebraic and geometric multiplicity, diagonalization, Cayley–Hamilton
7	Inner Products and Spectral Theory	Inner products, Cauchy–Schwarz, Gram–Schmidt, orthogonal projection and best approximation, spectral theorem (real symmetric), positive definiteness, Cholesky, Rayleigh quotient, Courant–Fischer
8	SVD, Norms, and Matrix Calculus	Quadratic forms, SVD theorem, Eckart–Young, pseudoinverse, operator and Frobenius norms, condition number, matrix exponential, gradient/Jacobian/Hessian, trace derivatives, chain rule

Table 1: Linear Algebra subdomain arc: Lessons 5–8.

Capstone summary: Linear Algebra subdomain (Lessons 5–8).

10 Further Reading

- **Phase 01, Functional Analysis** (Lessons on metric, normed, and Hilbert spaces) generalizes the finite-dimensional theory of this lesson to infinite dimensions. Inner products become Hilbert space inner products, the spectral theorem extends to compact self-adjoint operators, and the operator norm becomes the fundamental tool for analyzing bounded linear maps between Banach spaces.

- **Phase 01, Convex Analysis and Optimization** uses positive definiteness and matrix calculus extensively. The Hessian condition $H_f(x) \succ 0$ characterizes strict convexity, and gradient formulas derived here are the starting point for convergence analysis of first-order methods.
- Magnus and Neudecker [2019] is the definitive reference for matrix differential calculus, developing the theory via the *vec operator* and Kronecker products. Chapters 8–13 cover derivatives of eigenvalues, determinants, and matrix inverses.
- Petersen and Pedersen [2012] (the "Matrix Cookbook") provides a comprehensive catalog of matrix identities, derivatives, and decompositions organized for quick reference. It is an indispensable resource for implementing gradient computations in neural networks and RL algorithms.
- Horn and Johnson [2013, Chapters 5–7] develops the SVD, matrix norms, and perturbation theory (Weyl inequalities, Davis–Kahan theorem) in far greater depth, with applications to low-rank matrix completion and spectral clustering.
- **Historical note.** The singular value decomposition was independently discovered by Beltrami (1873) and Jordan (1874) for bilinear forms, and later placed in modern matrix form by Eckart and Young (1936). The name "singular values" was coined by Smithies (1937) in the context of integral equations. Matrix calculus was systematized by Magnus and Neudecker [2019] (first edition 1988), building on earlier work by Dwyer and Macphail (1948).

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